

Šematski prikaz raspodela diskretnog tipa

Raspodela	Prostor parametara	Zakon raspodele	$E(X)$	$D(X)$	Karakteristična funkcija
Bernulijeva	$0 < p < 1, q = 1 - p$	$p(k) = p^k q^{1-k}, k \in \{0, 1\}$	p	pq	$q + pe^{it}$
Binomna $X : \mathcal{B}(n, p)$	$0 < p < 1, q = 1 - p, n \in \mathbb{N}$	$p(k) = \binom{n}{k} p^k q^{n-k}, k \in \{0, 1, \dots, n\}$	np	npq	$(q + pe^{it})^n$
Poasonova $X : \mathcal{P}(\lambda)$	$\lambda > 0$	$p(k) = \frac{\lambda^k}{k!} e^{-\lambda}, k \in \{0, 1, 2, \dots\}$	λ	λ	$e^{-\lambda(1-e^{it})}$
Geometrijska $X : \mathcal{G}(p)$	$0 < p < 1, q = 1 - p$	$p(k) = q^{k-1} p, k \in \{1, 2, \dots\}$	$\frac{1}{p}$	$\frac{q}{p^2}$	$\frac{pe^{it}}{1 - qe^{it}}$

Šematski prikaz raspodela neprekidnog tipa

Raspodela	Funkcija gustine	Funkcija raspodele	$E(X)$	$D(X)$	Karakteristična funkcija
Uniformna $X : \mathcal{U}(a, b)$	$\varphi_X(x) = \begin{cases} \frac{1}{b-a}, & x \in (a, b) \\ 0, & x \notin (a, b) \end{cases}, a, b \in \mathbb{R}$	$F_X(x) = \begin{cases} 0, & x \leq a \\ \frac{x-a}{b-a}, & a < x \leq b \\ 1, & x > b \end{cases}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$\frac{e^{itb} - e^{ita}}{it(b-a)}$
Eksponencijalna $X : \mathcal{E}(a)$	$\varphi_X(x) = \begin{cases} 0, & x \leq 0 \\ ae^{-ax}, & x > 0 \end{cases}, a > 0$	$F_X(x) = \begin{cases} 0, & x \leq 0 \\ 1 - e^{-ax}, & x > 0 \end{cases}$	$\frac{1}{a}$	$\frac{1}{a^2}$	$\frac{a}{a - it}$
Normalna $X : \mathcal{N}(m, \sigma)$	$\varphi_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-m)^2}{2\sigma^2}}, x \in \mathbb{R}, m \in \mathbb{R}, \sigma \geq 0$	$F_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{(t-m)^2}{2\sigma^2}} dt, x \in \mathbb{R}$	m	σ^2	$e^{itm} e^{-\frac{\sigma^2 t^2}{2}}$